Core 1 Day 9 Hardware

Network Cables

Transmission Speeds
- ↳ Copper cables → 40 Gb upto
- ↳ Fiber cables → 100 Gb

Transmission Distance
- ↳ Copper Cables → 1100 meters (3609 feet)
- ↳ Fiber Cables → 40 Km (25 miles)

Analog telephone connector
↳ RJ 11

Twisted Pair Connector

RJ 11 : 4 pin connector, smaller

RJ45 : 8 pin connector
↳ found in computer networks, wider

Twisted Pair Categories

Category	Speed	Distance	Note
cat5	100Mbps	100 meters	older networks
cat5e	1000 Mbps/ 1Gbps	100meters	More twisted per foot
Cat 6	10000Mbps/ 10Gbps 1000 Mbps 1 Gbps	55meters 100 meters	Includes piece of plastic to minimize crosstalk.
Cat 6a	10,000 Mbps/ 10 Gbps	100 meters	Thicker wires

Cat7

Video Cables

HDMI (High Definition Multi-media Interface) → Support Audio Video both (HD & UHD)
- ↳ Standard (HDCP – High Definition Copy Protection)
- ↳ Mini
- ↳ Micro

Display Port cable

DVI port

VGA Connector
VGA port } – 15 pin

◦ Attenuation
is the loss of signal strength

◦ Noise Immunity
↳ EMI (Electro-Magnetic Interference)
↳ leak or disrupt signals

↳ Copper cables
↳ Highly susceptible to interference.
use shielded cables to protect against EMI

↳ Fiber Cables
are NOT susceptible to EMI since it is not Copper.

Wiring Standards (two)

↳ RJ45 pinouts
(T568 A and T568 B) standards
↳ The blue and brown pairs are always in the same location
↳ only orange and green pairs transpose their position

Fiber Optic Cables

not cheap

Advantages & Disadvantages

Fiber Optic Connector ☆☆
- ↳ ST connector (Straight Tip)
- ↳ LC connector (Lucent connection/ Local connection/ Little connector)

SMF (single mode fiber) vs MMF (multi mode fiber)

- ↳ SC connector (Standard connector/ Subscriber connector/ Square connector)
- ↳ Dual LC connector

Twisted Pair Cables ☆☆☆

↳ 8 cables inside
↓
4 pairs

most used cable
two types
- ↳ STP (Sheild Twisted Pair)
  - ↳ against EMI
  - ↳ ◦ Direct-burial cable
- ↳ UTP (Unshield Twisted Pair)
  - ↳ no protection against EMI

Coaxial Cable

↳ cable in or satellite television connections
◦ RG6 most common type

Advantage → Sheilded → EMI
→ Long transmission distance (1100 meters)
→ affordable than fiber optic

Disadvantage → More expensive than twisted pair
→ Copper Core can be snap if mishandled

Coax Connectors

BNC (old)

F connector

Cable	Connector	EMI Protection	Max Speed	Max distance	Conductor	Cost
Coax	BNC & F	Shielding	1 Gb	1100 meters	Copper	$$$
STP	RJ45 & RJ 11	Sheilding	10 Gb	100 meters	Copper	$$
UTP	RJ45 & RJ 11	None	10 Gb	100 meters	Copper	$
Single-mode Fiber	ST, SC & LC	NOT Susceptible	> 100Gb	40 Km	Glass	$$ $$ $$
Multi-mode Fiber	SC, LC	Not Suseptible	> 100 Gb	500 meters	Glass	$$$$$

Hard Drive Cables

↳ SATA (Serial ATA)
(Serial Advanced Technology Attachment)

SFF (Small Form Factor)

↳ SCSI
Older Drive Cable

↳ IDE / PATA